

PART II: CHEMISTRY

SECTION 1 (Maximum Marks: 18)

- This section contains SIX (06) questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from **0** to **9**, **BOTH INCLUSIVE**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer,
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct integer is entered;
Zero Marks : 0 If the question is unanswered;
Negative Marks : -1 In all other cases.

- *1. The 1st, 2nd, and the 3rd ionization enthalpies, I_1 , I_2 , and I_3 , of four atoms with atomic numbers n , $n + 1$, $n + 2$, and $n + 3$, where $n < 10$, are tabulated below. What is the value of n ?

Atomic Number	Ionization Enthalpy (kJ/mol)		
	I_1	I_2	I_3
n	1681	3374	6050
$n + 1$	2081	3952	6122
$n + 2$	496	4562	6910
$n + 3$	738	1451	7733

Sol.**9**

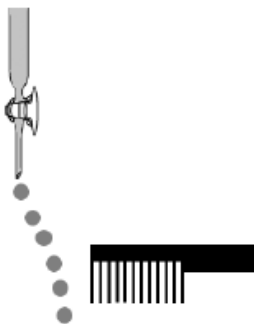
Since I.E. of Na is $495.8 \text{ kJ mol}^{-1}$, so $(n + 2)$ should be 11.
 $n + 2 = 11$, So $n = 9$

*2.

Consider the following compounds in the liquid form:

O_2 , HF , H_2O , NH_3 , H_2O_2 , CCl_4 , CHCl_3 , C_6H_6 , $\text{C}_6\text{H}_5\text{Cl}$.

When a charged comb is brought near their flowing stream, how many of them show deflection as per the following figure?

**Sol.****6**

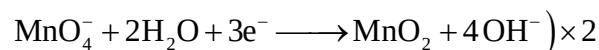
Only polar molecules can be attracted by a charged comb. So, except O_2 , CCl_4 and C_6H_6 all can be attracted in electric field.

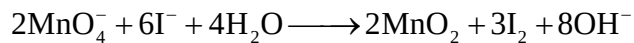
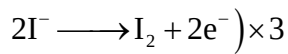
*3.

In the chemical reaction between stoichiometric quantities of KMnO_4 and KI in weakly basic solution, what is the number of moles of I_2 released for 4 moles of KMnO_4 consumed?

Sol.**6**

In slightly alkaline medium, MnO_4^- reduces to MnO_2 while I^- is oxidized to I_2 .



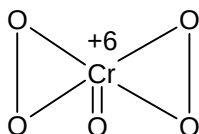
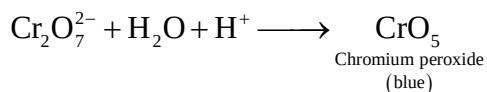
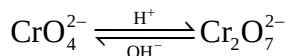


So, 2 mol of KMnO_4 can produce 3 mole of I_2

$\therefore$ 4 mol of KMnO_4 can produce 6 mole of I_2

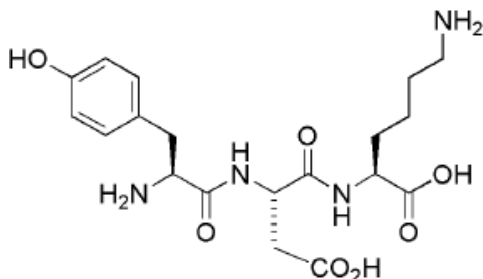
- *4. An acidified solution of potassium chromate was layered with an equal volume of amyl alcohol. When it was shaken after the addition of 1 mL of 3% H_2O_2 , a blue alcohol layer was obtained. The blue color is due to the formation of a chromium (VI) compound 'X'. What is the number of oxygen atoms bonded to chromium through only single bonds in a molecule of X?

Sol. 4



So, number of oxygen atoms attached to Cr through single bond = 4

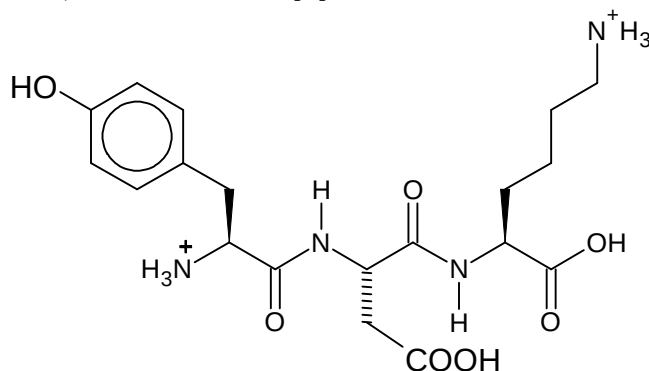
5. The structure of a peptide is given below:



If the absolute values of the net charge of the peptide at pH = 2, pH = 6, and pH = 11 are $|z_1|$, $|z_2|$, and $|z_3|$, respectively, then what is $|z_1| + |z_2| + |z_3|$?

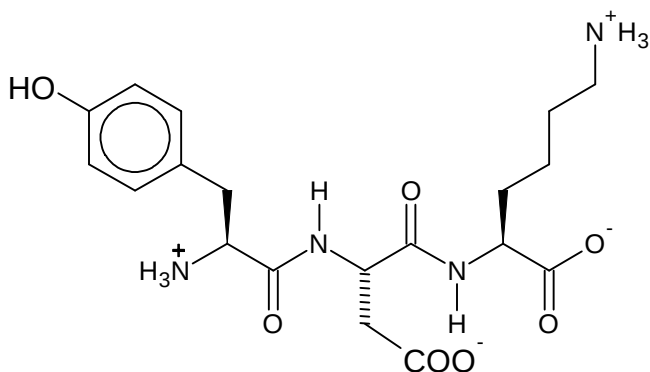
Sol. 5

At pH = 2 (highly acidic), the structure of the peptide will be as shown below.



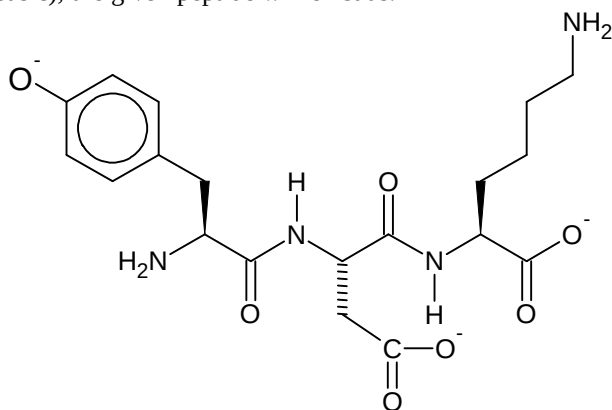
So, $|z_1| = 2$

At pH = 6 it will exist as neutral or zwitter-ion structure as shown below.



So, according to this condition, $|Z_2| = 0$

At pH = 11 (highly basic), the given peptide will exist as:



So, $|Z_3| = |-3| = 3$

$|Z_1| + |Z_2| + |Z_3| = 2 + 0 + 3 = 5$

- *6. An organic compound ($C_8H_{10}O_2$) rotates plane-polarized light. It produces pink color with neutral $FeCl_3$ solution. What is the total number of all the possible isomers for this compound?

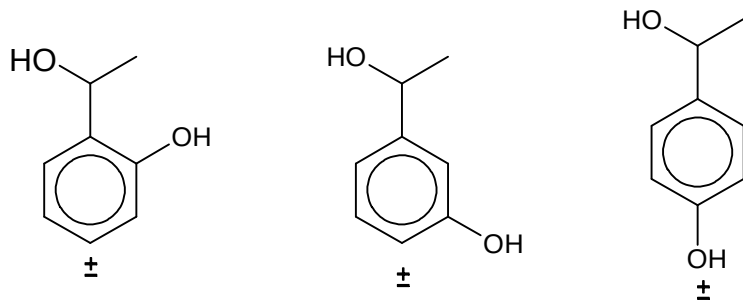
Sol. 6 (Considering that all isomers satisfy the given conditions)

Since given compound produces pink colour with neutral $FeCl_3$.

So, it must have at least one $-OH$ group attached to benzene ring.

Also, given compound is optically active, it means it should have a chiral centre, which is only possible when it has $-CH(OH)CH_3$

So, based on the above observations, compound ($C_8H_{10}O_2$) is expected to possess the following isomeric structures.



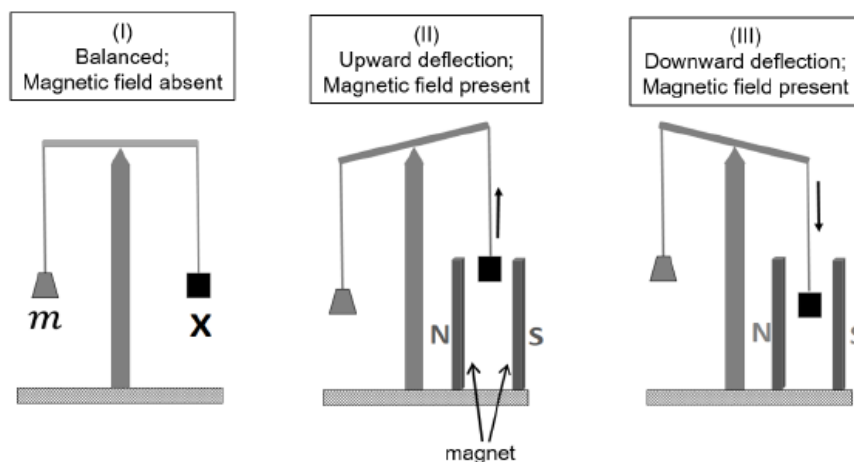
So, total isomeric structures = 6

SECTION 2 (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme.

Full Marks	:	+4	If only (all) the correct option(s) is(are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY two options are chosen,
Partial Marks	:	+2	If three or more options are correct but ONLY one options are chosen, both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e., the question is unanswered);
Negative Marks	:	-2	In all other cases.

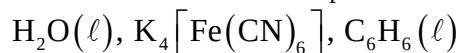
- *7. In an experiment, m grams of a compound **X** (gas/liquid/solid) taken in a container is loaded in a balance as shown in figure **I** below. In the presence of a magnetic filed, the pan with **X** is either deflected upwards (figure **II**), or deflected downwards (figure **III**), depending on the compound **X**. Identify the correct statements(s).



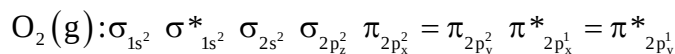
- (A) If **X** is $\text{H}_2\text{O}(l)$, deflection of the pan is upwards.
 (B) If **X** is $\text{K}_4[\text{Fe}(\text{CN})_6]$ (s), deflection of the pan is upwards.
 (C) If **X** is $\text{O}_2(g)$, deflection of the pan is downwards.
 (D) If **X** is $\text{C}_6\text{H}_6(l)$, deflection of the pan is downwards.

Sol. **A,B,C**

Diamagnetic (all electrons paired) are repelled i.e. magnetized in opposite direction in magnetic field, so such substance are deflected upwards in magnetic field. Diamagnetic substances are



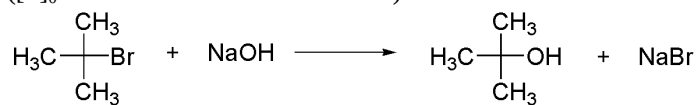
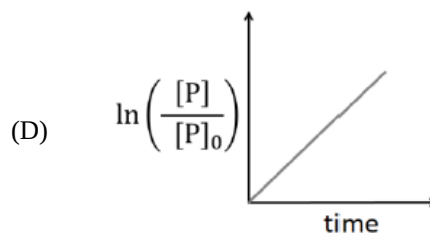
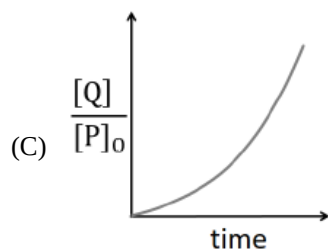
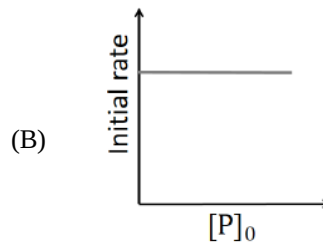
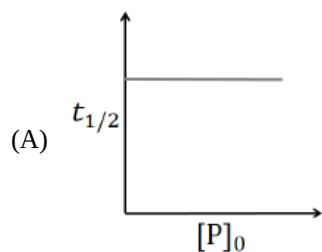
Paramagnetic substances (at least one unpaired electron) are attracted in magnetic field, so such substances are deflected downwards.



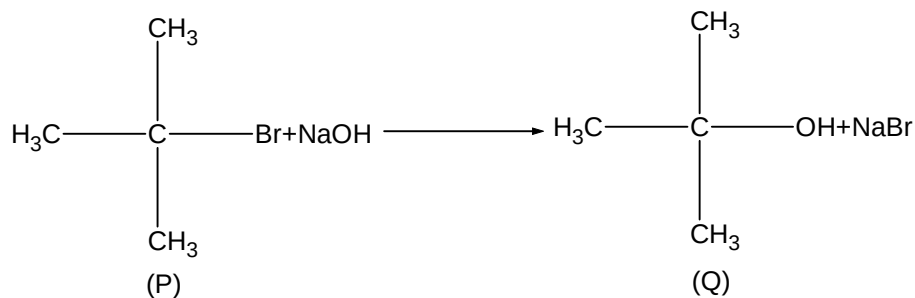
Since O_2 molecules has two unpaired electrons in antibonding π . MO.

So, O_2 is paramagnetic.

8. Which of the following plots is (are) correct for the given reaction?
 ($[P]_0$ is the initial concentration of **P**)

**P****Q**

Sol. A



The order of the above reaction is one (S_N1) because substrate is a 3° -halide

So, for a first order reaction.

$$P = P_0 \times e^{-kt} \quad \text{or} \quad \frac{P}{P_0} = e^{-kt}$$

$$\ln\left(\frac{P}{P_0}\right) = -kt$$

$$y = mx$$

So, the slope of the graph $\ln\left(\frac{P}{P_0}\right)$ versus time (t) should be negative.

Hence option D is incorrect.

Also $\frac{P}{P_0} = e^{-kt}$

$$\left(1 - \frac{P}{P_0}\right) = (1 - e^{-kt})$$

$$\frac{P_0 - P}{P_0} = (1 - e^{-kt})$$

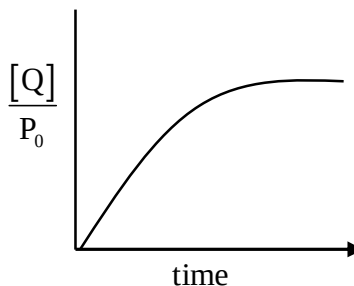


Fig:1

Or $\frac{[Q]}{P_0} = (1 - e^{-kt})$

So, graph between $\frac{[Q]}{P_0}$ Vs. t , should be as shown in figure (1), so, option (C) is incorrect

For a first order reaction

$$r = k[P]^1$$

At $t = 0$, $r_i = k[P_0]$

So, initial rate of reaction is directly proportional to initial conc. of P.

Hence option B is also incorrect.

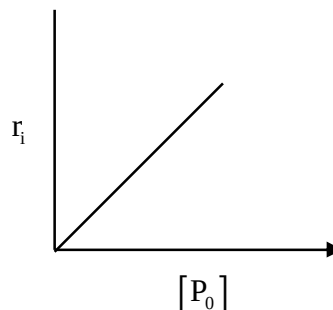


Fig:2

For a first order reaction

$$t_{1/2} = \frac{\ln 2}{k} [P_0]^0$$

So, $t_{1/2}$ is independent of initial conc. of reactant.

So, option (A) is correct.

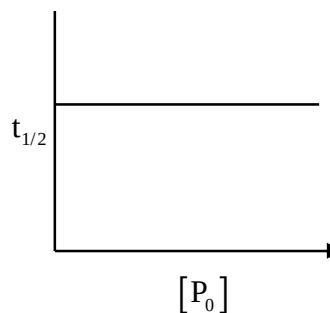
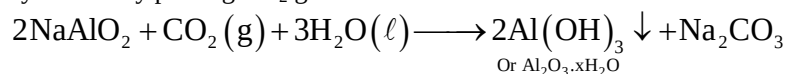


Fig:3

9. Which among the following statement(s) is(are) true for the extraction of aluminium from bauxite?
- (A) Hydrated Al_2O_3 precipitates, when CO_2 is bubbled through a solution of sodium aluminate.
- (B) Addition of Na_3AlF_6 lowers the melting point of alumina.
- (C) CO_2 is evolved at the anode during electrolysis.
- (D) The cathode is a steel vessel with a lining of carbon.

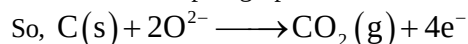
Sol. A,B,C,D

Sodium meta aluminate is precipitated in the form of $Al(OH)_3$ or $Al_2O_3 \cdot xH_2O$. i.e hydrated aluminium hydroxide by passing CO_2 gas.



In electrolytic reduction of Al_2O_3 by Hall-Haroult process, cryolite (Na_3AlF_6) and fluorspar (CaF_2) are added in order to reduce the melting temperature of Al_2O_3 .

Anode is made up of graphite,



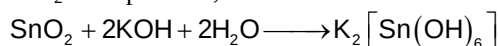
Cathode is a steel vessel lined with graphite.

10. Choose the correct statement(s) among the following
- (A) $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ is a reducing agent.
 (B) SnO_2 reacts with KOH to form $\text{K}_2[\text{Sn}(\text{OH})_6]$.
 (C) A solution of PbCl_2 in HCl contains Pb^{2+} and Cl^- ions.
 (D) The reaction of Pb_3O_4 with hot dilute nitric acid to give PbO_2 is a redox reaction.

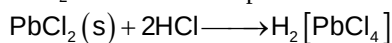
Sol. A, B

(A) Sn^{2+} is less stable than Sn^{4+} , therefore $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ is a Lewis acid.

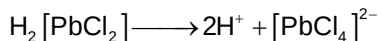
(B) SnO_2 is amphoteric, so it will dissolve in KOH solution forming K_2SnO_3 or $\text{K}_2[\text{Sn}(\text{OH})_6]$.



(C) PbCl_2 will form a complex with HCl

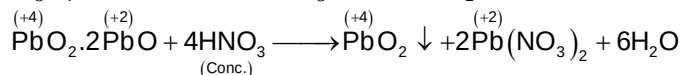


So, we may assume that now this complex will dissociate as:



So, considering this possibility, option (C) should be incorrect.

(D) Pb_3O_4 reacts with conc. HNO_3 to form PbO_2 but the reaction is not a redox reaction.

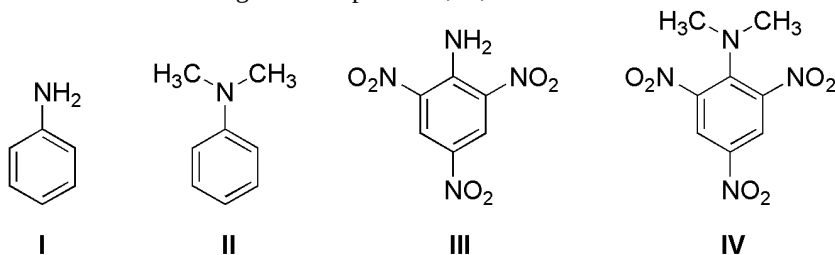


or



So, option (D) is incorrect.

- *11. Consider the following four compounds **I**, **II**, **III** and **IV**.



Choose the correct statement(s).

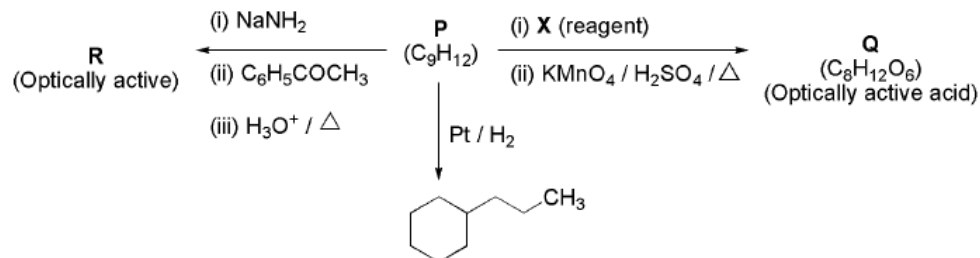
- (A) The order of basicity is **II** > **I** > **III** > **IV**.
 (B) The magnitude of pK_b difference between **I** and **II** is more than that between **III** and **IV**.
 (C) Resonance effect is more in **III** than in **IV**.
 (D) Steric effect makes compound **IV** more basic than **III**.

Sol. C, D

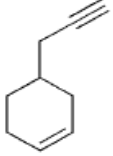
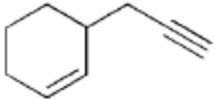
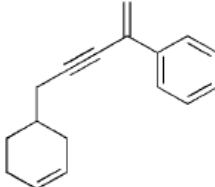
- ❖ (III) is the weakest base due to the presence of three $-\text{NO}_2$ group, which exert strong $-M$ effect.
- ❖ In compound (IV) $-\text{N}(\text{Me})_2$ group is thrown out of the plane of aromatic ring due to steric crowding by $-\text{NO}_2$ groups at ortho positions as a result of this lone pair of electron on N – atom does not participate in resonance with aromatic ring and hence it acts as an aliphatic amine. So, (IV) is the strongest base.
- ❖ Correct order of basicity is; **III** < **I** < **II** < **IV** (in basic character).

- ❖ Difference of pK_a values is highest between (III) and (IV).
- ❖ pK_b ; (I) $\rightarrow$ 9.42; (II) $\rightarrow$ 8.94, (IV) $\rightarrow$ 4.60

12. Consider the following transformations of a compound **P**.

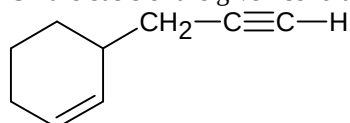


Choose the correct option(s).

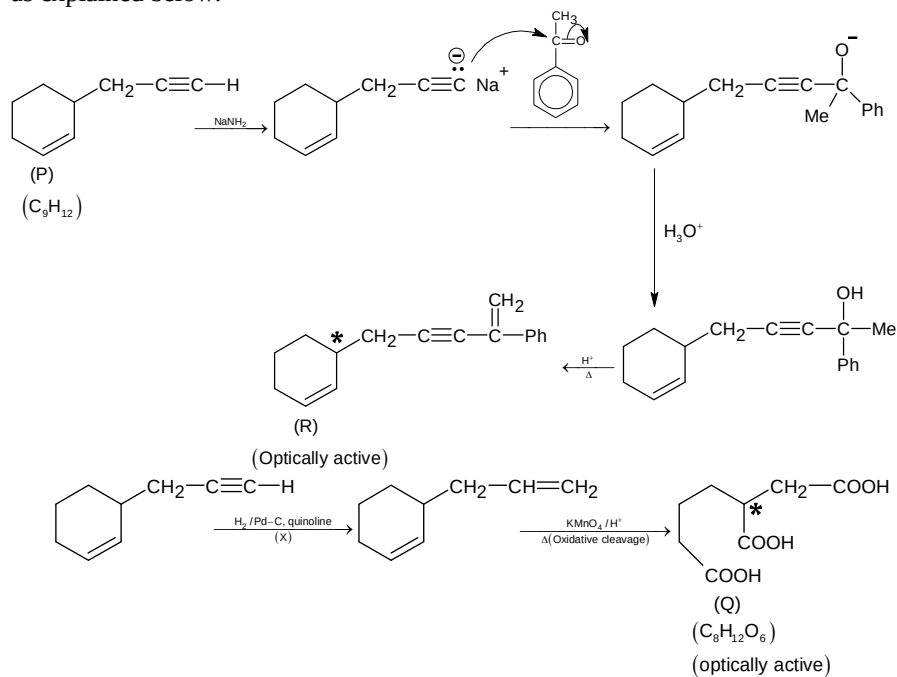
- (A) **P** is 
- (B) **X** is Pd-C/quinoline/H₂
- (C) **P** is 
- (D) **R** is 

Sol. B, C

- ❖ On the basis of the given conditions, correct structure of (**P**) should be



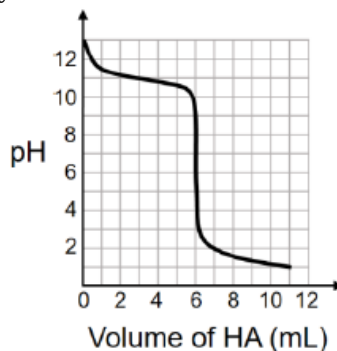
as explained below:



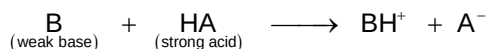
SECTION 2 (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer, If the numerical value has more than two decimal places, truncate/round-off the value of **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme.
Full Marks : +4 If ONLY the correct numerical value is entered;
Zero Marks : 0 In all other cases.

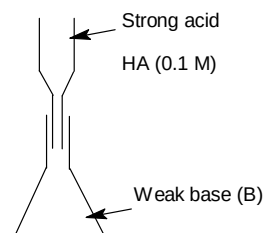
- *13. A solution of 0.1 M weak base (B) is titrated with 0.1 M of a strong acid (HA). The variation of pH of the solution with the volume of HA added is shown in the figure below. What is the pK_b of the base? The neutralization reaction is given by $B + HA \rightarrow BH^+ + A^-$.



Sol. 03.00



- ❖ From the graph, it is clear, that the equivalence point is reached at a titre value of 6 mL i.e. when 6 mL of HA are added base (B) is completely neutralized. So it will be half neutralized at titre value of 3 mL; when 3 mL of HA is added, 'B' is half – neutralized. So at this stage it will form a best buffer.
 i.e. $pOH = pK_b = 3$
 So, pK_b of weak base = 3



14. Liquid **A** and **B** form ideal solution for all compositions of **A** and **B** at 25°C . Two such solutions with 0.25 and 0.50 mole fractions of **A** have the total vapour pressures of 0.3 and 0.4 bar, respectively. What is the vapour pressure of pure liquid **B** in bar?

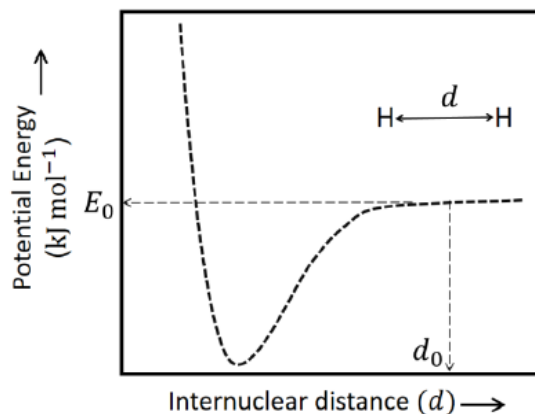
Sol. 0.20

$$0.3 = p_A^0 \times 0.25 + p_B^0 \times 0.75 \quad \dots(1)$$

$$0.4 = p_A^0 \times 0.50 + p_B^0 \times 0.50 \quad \dots(2)$$

On solving, $p_B^0 = 0.20$ bar and $p_A^0 = 0.6$ bar

- *15. The figure below is the plot of potential energy versus internuclear distance (d) of H_2 molecule in the electronic ground state. What is the value of the net potential energy E_0 (as indicated in figure) in kJ mol^{-1} , for $d = d_0$ at which the electron-electron repulsion and the nucleus-nucleus repulsion energies are absent? As reference, the potential energy of H atom is taken as zero when its electron and the nucleus are infinitely far apart.
Use Avogadro constant as $6.023 \times 10^{23} \text{ mol}^{-1}$.



Sol. **-5246.50**

$$\text{P.E.} = -\frac{Kq_1q_2}{r}$$

$$\text{P.E. of one H - atom in ground state} = -\frac{K(e)(e)}{r}$$

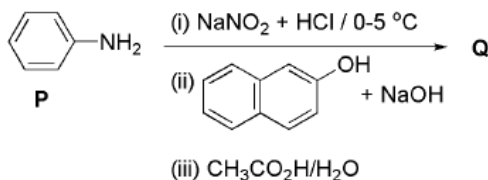
$$\text{P.E.} = \frac{-9 \times 10^9 \times (1.6 \times 10^{-19})^2}{0.529 \times 10^{-10}} = -43.5538 \times 10^{-19} \text{ J/atom}$$

So, P.E. of two H - atoms (or one molecules of H_2) = $-2 \times 43.5538 \times 10^{-19} = -87.106 \times 10^{-19} \text{ J/molecule}$

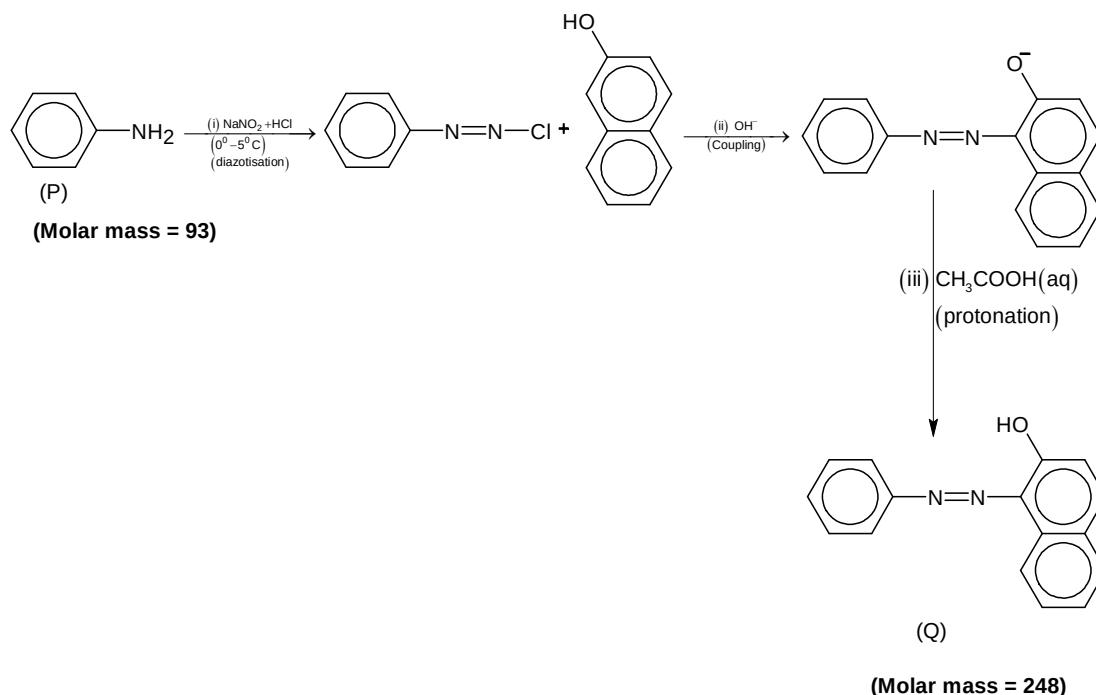
So, P.E. of one mole of H_2 molecules = $-87.106 \times 10^{-19} \times 6.023 \times 10^{23} \text{ J}$

$$\begin{aligned} &= -524.6490 \times 10^4 \\ &= -5246.49 \text{ kJ/mol}^{-1} \\ &\approx -5246.50 \text{ kJmol}^{-1} \end{aligned}$$

16. Consider the reaction sequence from **P** to **Q** shown below. The overall yield of the major product **Q** from **P** is 75%. What is the amount in grams of **Q** obtained from 9.3 mL of **P**? (Use density of **P** = 1.00 g mL^{-1} ; Molar mass of C = 12.0, H = 1.0, O = 16.0 and N = 14.0 g mol^{-1})



Sol. **18.60**



$$\text{Moles of P} = \frac{9.3}{93} = 0.1 = \text{moles of Q.}$$

$$\text{So, mass of (Q)} = 0.1 \times 248 \times 0.75 = 18.60 \text{ gram}$$

- *17. Tin is obtained from cassiterite by reduction with coke. Use the data given below to determine the minimum temperature (in K) at which the reduction of cassiterite by coke would take place.

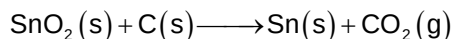
$$\text{At } 298 \text{ K; } \Delta_f H^\circ (\text{SnO}_2(\text{s})) = -581.0 \text{ kJ mol}^{-1}, \Delta_f H^\circ (\text{CO}_2(\text{g})) = -394.0 \text{ kJ mol}^{-1},$$

$$S^\circ (\text{SnO}_2(\text{s})) = 56.0 \text{ J K}^{-1} \text{ mol}^{-1}, S^\circ (\text{Sn}(\text{s})) = 52.0 \text{ J K}^{-1} \text{ mol}^{-1},$$

$$S^\circ (\text{C}(\text{s})) = 6.0 \text{ J K}^{-1} \text{ mol}^{-1}, S^\circ (\text{CO}_2(\text{g})) = 210.0 \text{ J K}^{-1} \text{ mol}^{-1},$$

Assume that the enthalpies and the entropies are temperature independent.

Sol. 935



$$\Delta H_r^\circ = -394.0 + 581 = 187 \text{ kJ mol}^{-1}$$

$$\Delta S^\circ = 210 + 52 - 6 - 56 = 200 \text{ J mol}^{-1} \text{ K}^{-1}$$

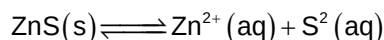
$$\text{Equilibrium temperature } (T)_{\text{eq}} = \frac{187 \times 1000}{200} = 935 \text{ K}$$

So, $T > 935 \text{ K}$ (for spontaneity)

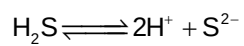
- *18. An acidified solution of 0.05 M Zn^{2+} is saturated with $0.1 \text{ M H}_2\text{S}$. What is the minimum molar concentration (M) of H^+ required to prevent the precipitation of ZnS ?

$$\text{Use } K_{\text{sp}} (\text{ZnS}) = 1.25 \times 10^{-22} \text{ and overall dissociation constant of } \text{H}_2\text{S. } K_{\text{NET}} = K_1 K_2 = 1 \times 10^{-21}$$

Sol. 0.20



$$[S^{2-}] = \frac{K_{sp}}{[Zn^{2+}]} = \frac{1.25 \times 10^{-22}}{0.05} = 25 \times 10^{-22}$$



$$10^{-21} = \frac{[H^+]^2 \times 25 \times 10^{-22}}{0.1}$$

$$[H^+]^2 = \frac{10^{-22} \times 10^{22}}{25} = \frac{1}{25}$$

$$[H^+] = \frac{1}{5} = 0.20 \text{ M}$$